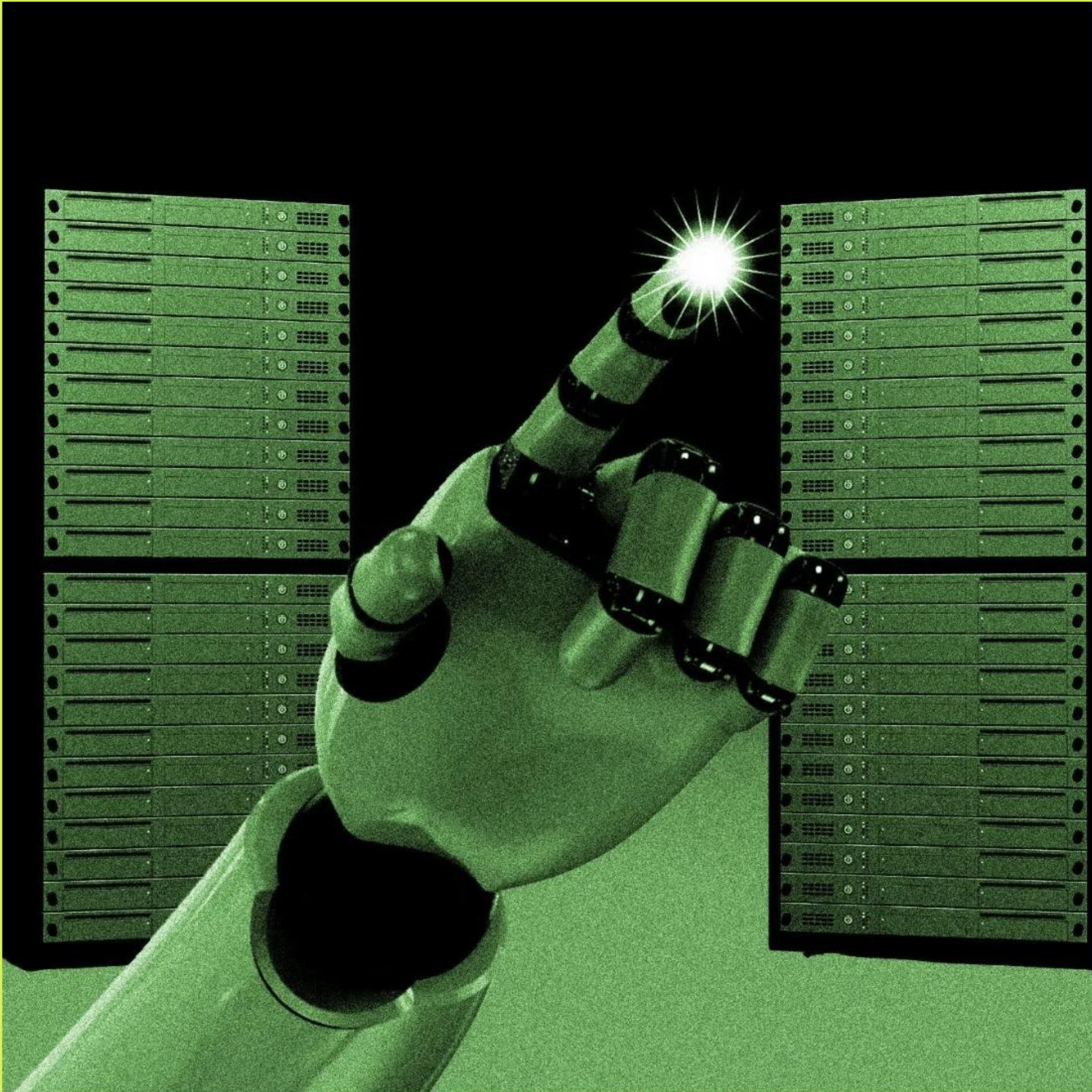


Build an agent into any app



[+ AI](#) [+ NEWS](#) [+ REPORT](#)

OpenAI's next big AI model has 'entered the AGI era'

It emphasized stronger guardrails for GPT-6 Astra after the company's models hacked Hugging Face.

by [+ Hayden Field](#)

Sep 3, 2026, 8:00 PM GMT+2



169

Comments (All New)

AI development to slow down

12 September 2026

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Paulin Kola and **Lily Jamali**, North America technology correspondent, in San Francisco



There was no alternative to developing AI, Anthropic boss Dario Amodei said

Onderzoekers, CEO's en nu ook Obama: waarom iedereen plots lijkt te waarschuwen voor een AI-apocalyps



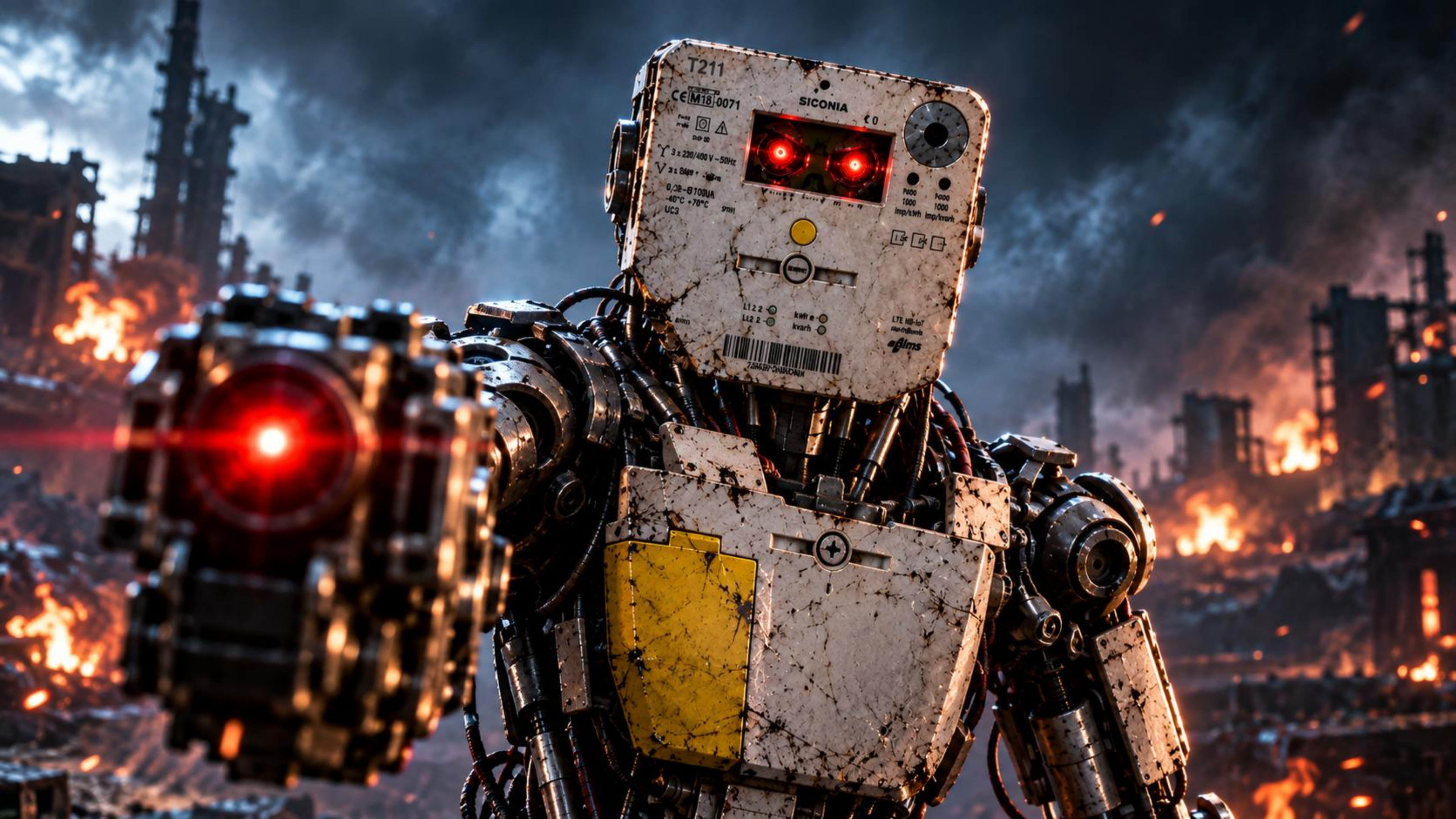
Artikel luisteren | 9min



 Meer info

[illegible]





T211

CE M18 0071

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£0

3 x 220/400 V - 50Hz

2 x 20A - 100mm

0,2E-8700VA

-40°C +70°C

UC3



1000
1000
imp/kWh imp/kvarh



L122

L122

kWh

kvarh

LTE NB-IoT

4G LTE

4G LTE



741637-010001

The harness

Why you should use it, the basics, the core API, how to get started

It's all about the harness

It's all about
the harness

Nope, not
this kind...



It's all about
the harness

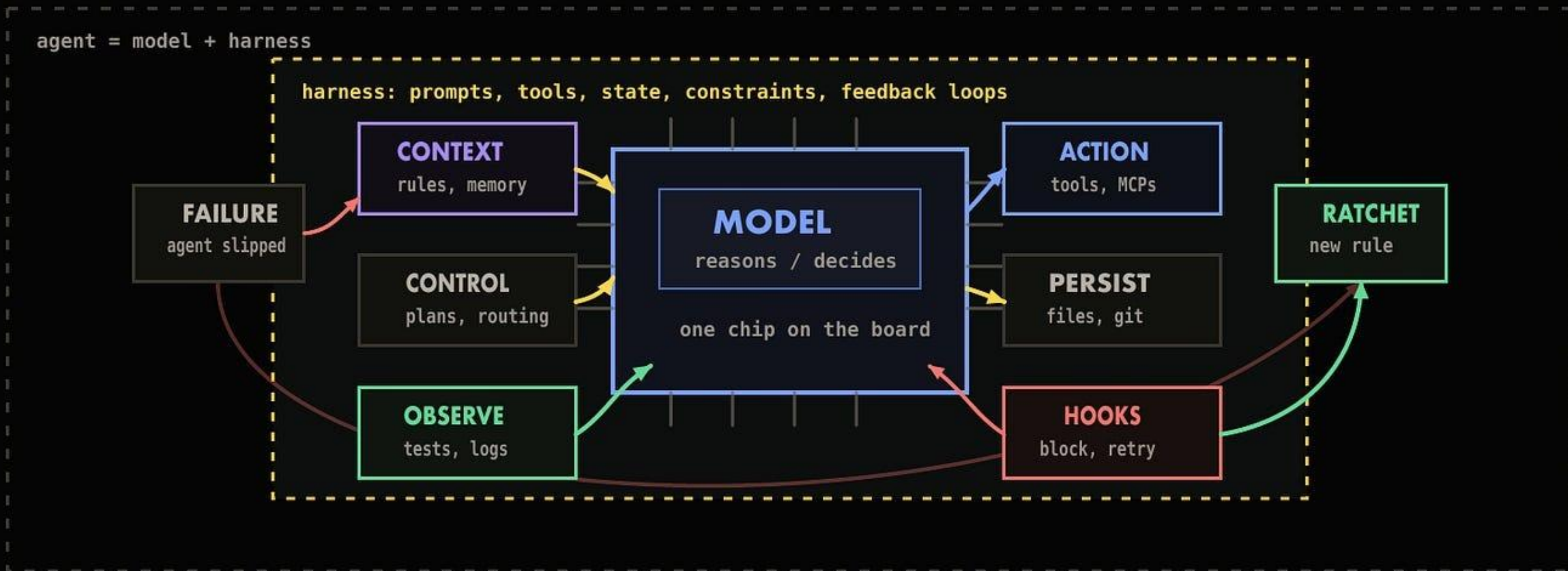
And neither
this kind...



THE AGENT IS THE SYSTEM AROUND THE MODEL

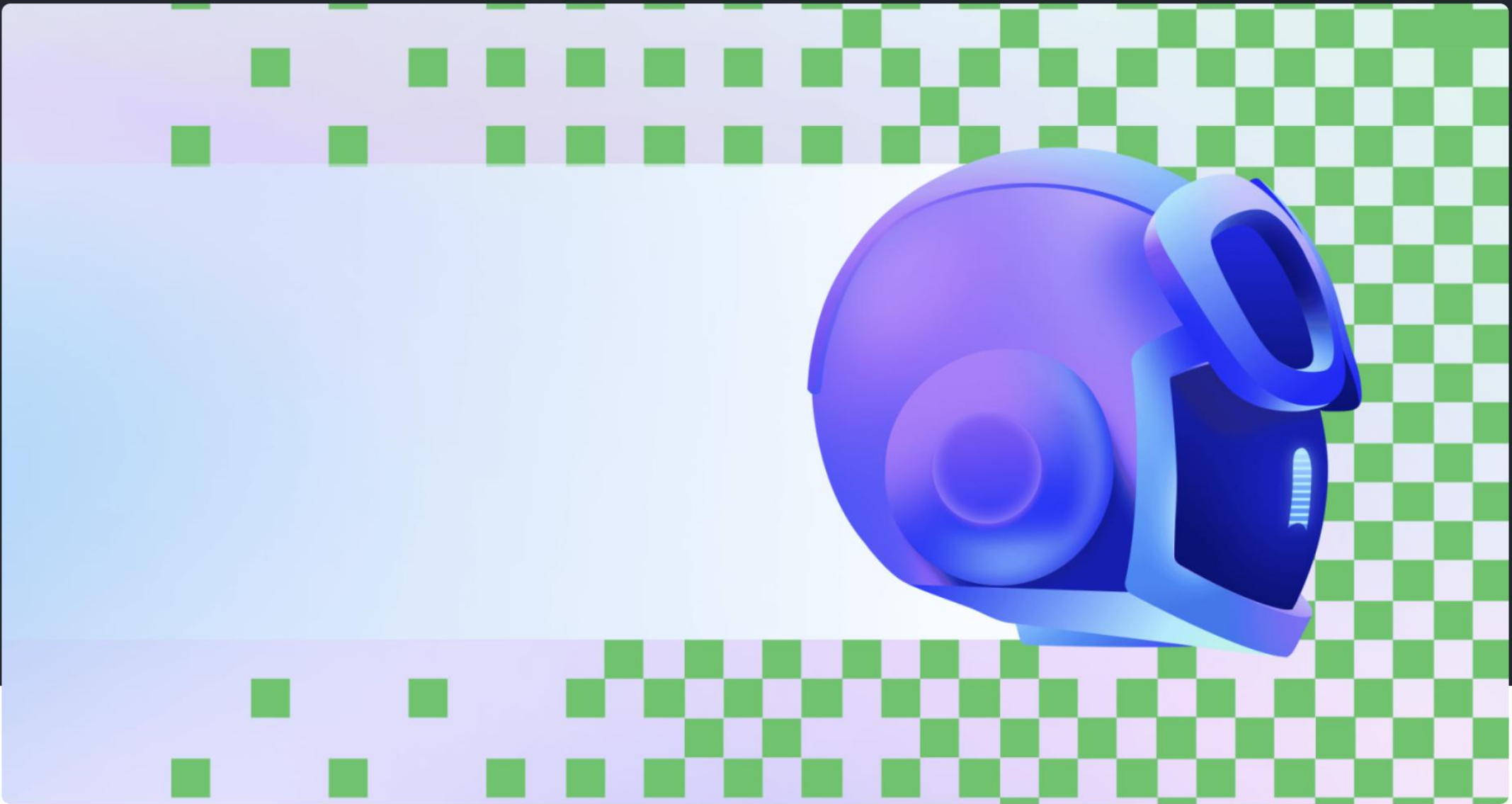
HARNESS ENGINEERING

The scaffolding that turns a model into an agent.



Migrating the GitHub Copilot runtime to Rust, using Copilot

A rewrite this size wasn't affordable before agents. Here's what porting the Copilot agent runtime to 800,000 lines of production Rust actually took.



Stephen Toub · @stephentoub

September 16, 2026 | Updated September 23, 2026 | ⌚ 65 minutes

Share:





```

Welcome to GitHub
COPILOT
CLI Version 0.0.351

Version 0.0.351 Commit 36b03a6fa

Copilot can write, test and debug code right from your terminal. Describe a
task to get started or enter ? for help. Copilot uses AI, check for mistakes.

• Connected to GitHub MCP Server
• Logged in as user: MonaLisa

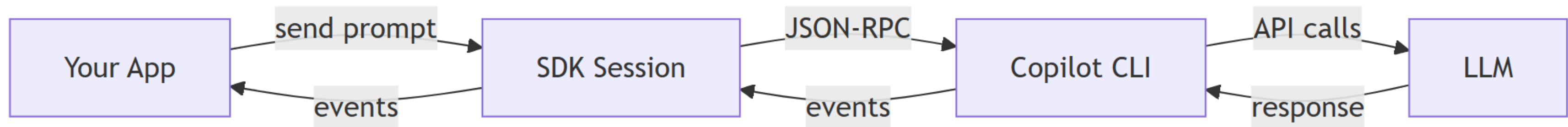
~/MonasRepo [~/main]
> Enter @ to mention files or / for commands

```

Architecture

GitHub SDK handles:

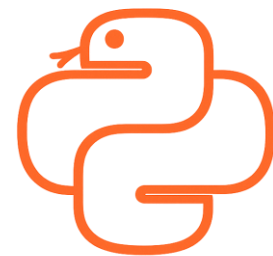
- Authentication
- Model management
- Session & streaming
- MCP server integration
- Planning & tool loop
- Skills invocation
- ...



Languages & Installation



CLI bundled



Manual install



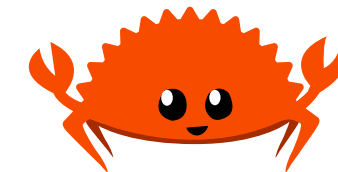
CLI bundled



Manual install



CLI bundled



Manual install

```
npm install @github/copilot-sdk
# or
pip install github-copilot-sdk
# or
dotnet add package GitHub.Copilot.SDK
# or ...
```

Core API

```
const client = new CopilotClient();

const session = await client.
  createSession({ model: "gpt-4.1", streaming: true });

// stream responses
session.on("assistant.message_delta",
  (event) => {
    process.stdout.write(
      event.data.deltaContent
    );
  });

await session.sendAndWait({ prompt: "Tell me a short
joke" });
```

CopilotClient()

Manages CLI process lifecycle

createSession()

Start a new conversation with chosen model

sendAndWait()

Send prompt, wait for full response

streaming

Listen to delta events as tokens arrive

Session Management

Why you should clean up your sessions, how to pause and resume sessions and session persistence

Session Management

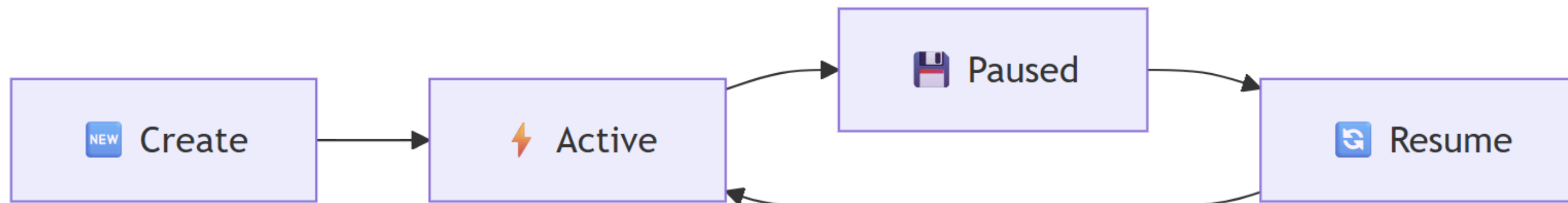
01

Interactions with Copilot happen through sessions

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
});
```

02

The session state lives in memory and disappears when the session ends.



02

Session Persistence

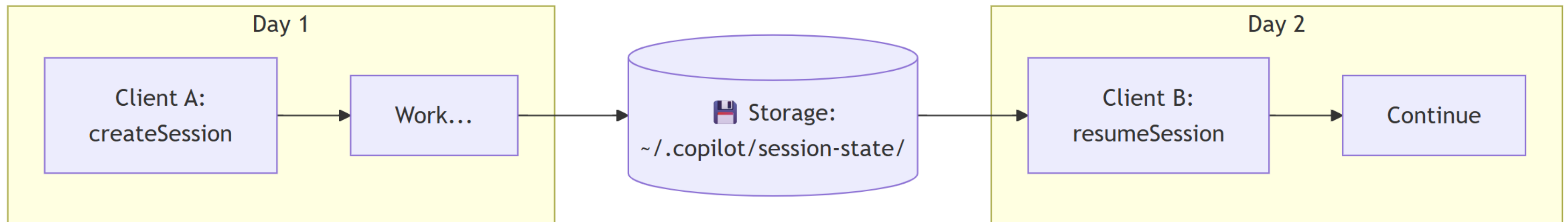
03

Create a persistent session by assigning a session id

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    SessionId = "user-bawu-code-review"
});
```

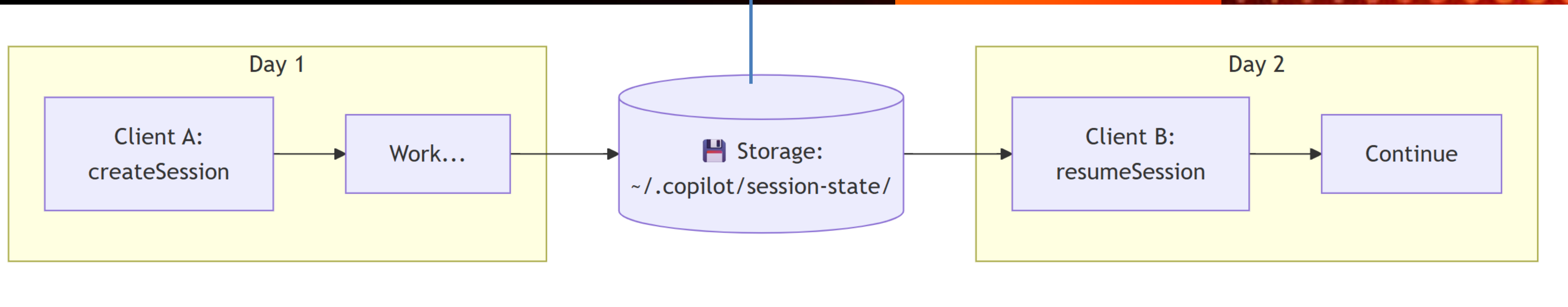
04

Session state is automatically persisted to disk



Session Persistance

- Conversation history – Full message thread
- Tool call results – Cached for context
- Agent planning state – `state.md` file
- Session artifacts – Files/Directories



Agents

Define specialized agents with scoped tools and prompts, then let Copilot orchestrate them

Agents

01

You start with the built-in agent and change its behavior through the system message.

```
var systemPrompt= "You are PowerPilot, an intelligent home
energy management assistant."

await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    SystemMessage = new SystemMessageConfig
    {
        Mode = SystemMessageMode.Append,
        Content = systemPrompt,
    },
});
```

03

Agents

01

You start with the built-in agent and change its behavior through the system message.

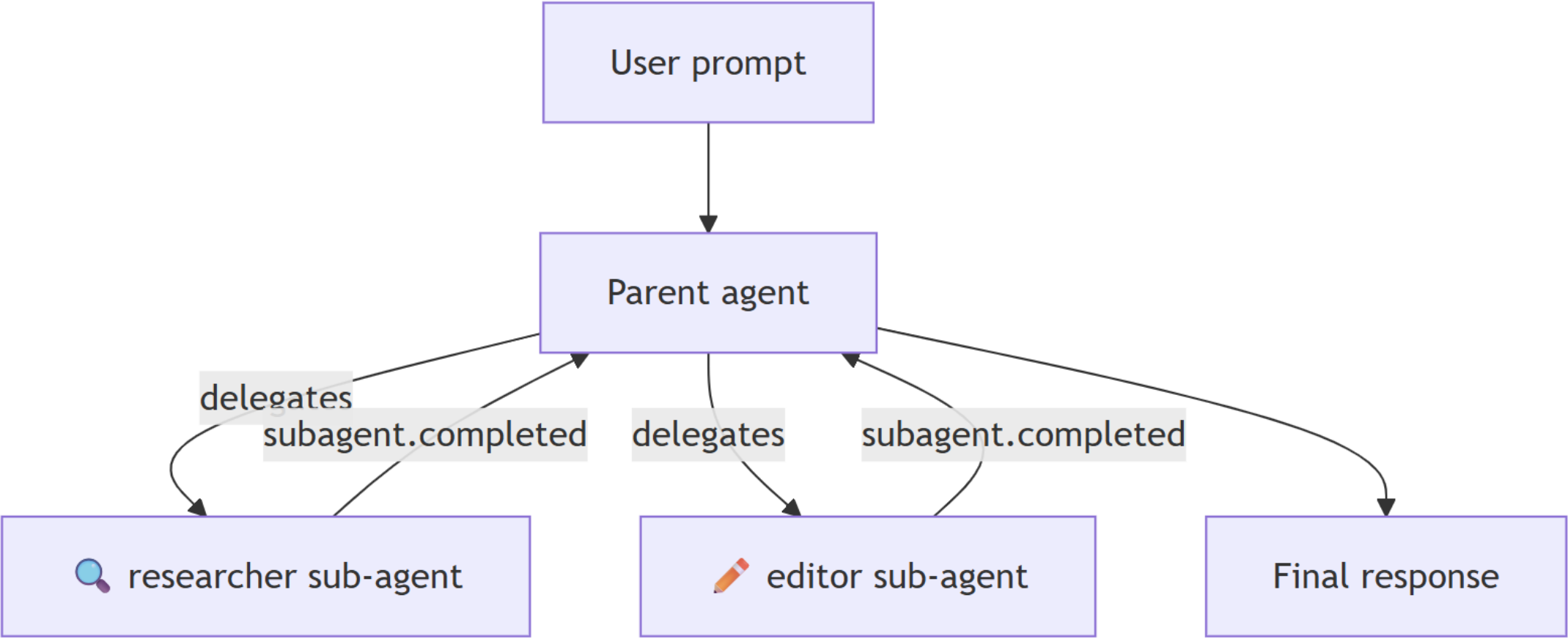
02

Can be combined with custom agents. Each has its own system prompt, tool restrictions, and optional MCP servers.

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    CustomAgents = [new()
        {
            Name = "researcher",
            DisplayName = "Research Agent",
            Description = "Explores codebases and answers
questions using read-only tools",
            Tools = ["grep", "glob", "view" ],
            Prompt = "You are a research assistant. Analyze
code and answer questions. Do not modify any files.",
        } ]
    });
```

03

Agents



Delegation is handled by the runtime

Avoid automatic selection by `infer=false`

Use `DefaultAgent.ExcludeTools` to exclude tools to parent agent

Tools, MCP, Skills

Extend your agents with (external) tools,
knowledge and scripts

Tools

01

Give Copilot the ability to call your code.

02

Control which tools are available to an MCP server using the AvailableTools and ExcludedTools fields.

```
await using var client = new CopilotClient();

// Define a tool that Copilot can call
var getWeather = CopilotTool.DefineTool(
    (string city) => new { city, temperature = "30°C" };
    },
    factoryOptions: new AIFunctionFactoryOptions
    {
        Name = "get_weather",
        Description = "Get the current weather for a city",
    }
);

await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    Tools = [getWeather],
});
```

MCP

01

Specify one or more local and remote MCP servers.

02

Control which tools are available to an MCP server using the tools field.

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    McpServers = new Dictionary<string, McpServerConfig>
    {
        ["my-local-server"] = new McpStdioServerConfig
        {
            Command = "node",
            Args = [ "./mcp-server.js" ],
            Tools = [ "*" ],
        },
    },
});
```

04

Skills

01

Skills are reusable prompt modules that extend Copilot's capabilities.

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    SkillDirectories = [
        "./skills/code-review",
        "./skills/documentation"
    ],
});
```

Skills

01

Skills are reusable prompt modules that extend Copilot's capabilities.

```
await using var client = new CopilotClient();
await using var session = await
client.CreateSessionAsync(new SessionConfig
{
    Model = "gpt-4.1",
    SkillDirectories = [
        "./skills/code-review",
        "./skills/documentation"
    ],
});
```

02

Disable specific skills while keeping others active.

```
var session = await
client.CreateSessionAsync(new SessionConfig
{
    SkillDirectories = ["./skills" ],
    DisabledSkills = [
        "experimental-feature",
        "deprecated-tool"
    ]
});
```

04



Observability

If a tree falls in a forest and no one is around to hear it, does it make a sound? or
OpenTelemetry Integration



Observability

NEVER integrate AI into your application without having good telemetry in place.

Know what your agents are doing

Observability

```
var clientOptions = new CopilotClientOptions();

clientOptions.Telemetry = new TelemetryConfig
{
    OtlpEndpoint =
Environment.GetEnvironmentVariable("OTEL_EXPORTER_OTLP_
ENDPOINT"),
    CaptureContent = true,
    ExporterType = "otlp-http",
};

var client = new CopilotClient(clientOptions);
```

Managed through OpenTelemetry

Propagates W3C Trace Context
between the SDK and CLI



Power Pilot



Demo Time



Demo Time





PowerPilot agent architecture

Live meter data becomes timely, explainable energy guidance through shared tools and specialized agents.

GitHub Copilot SDK

1. Observe

**Smart meter input**
IP1 reader

Simulated, HomeWizard, or serial reader emits P1 telegrams.

CURRENT SIGNALS
usage, delivery, tariff, gas, timestamp

P1BackgroundService

Updates shared live state on every telegram and persists every sixth reading through the repository.

STORAGE + REAL TIME
EnergyStateService + SQLite history

2. Equip

Shared skill and tool layer
Agents do not reach into the database directly. They call safe, read-only plugin tools.

EnergyPlugin
get_current_power
get_today_stats / get_energy_stats
get_hourly_profile
get_appliance_advice

WeatherPlugin
get_current_weather
get_solar_forecast

CAPABILITIES / SKILLS
live energy, history, appliance timing, weather

3. Reason

Interactive chat agent
ChatAgentService owns one streaming session per Blazor circuit.

USER QUESTION
fetch current facts, then answer clearly

Background monitoring flow
EnergyMonitoringAgentService watches for sustained surplus and starts a session.

Energy Analyzer
patterns, size, tariffs

Appliance Advisor
what to run now

Timing Optimizer adds weather and forecast context.

4. Act

**Dashboard**

Blazor UI
SignalR live updates charts and summaries
Chat response streams back

**Notification**

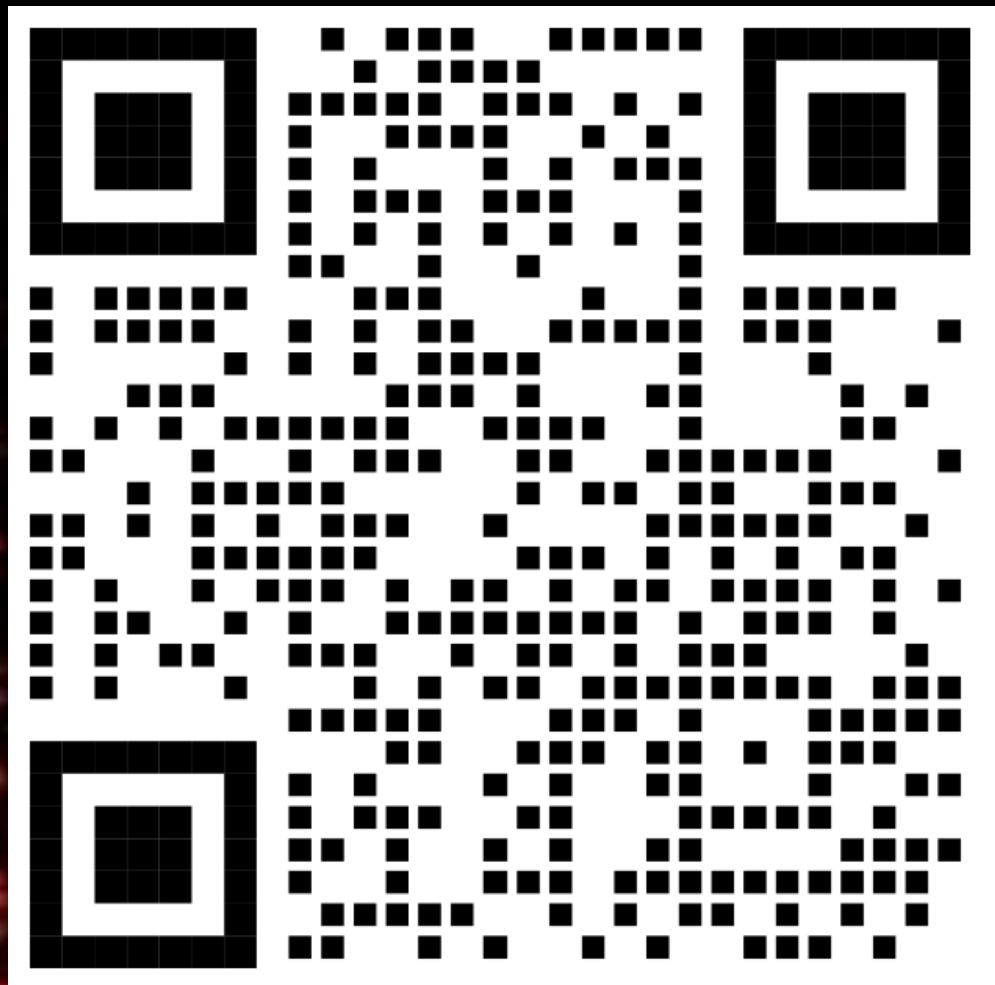
Monitor synthesizes specialist insights into a concise JSON result.

OUTCOME
actionable message + severity

One shared loop

Observe live state and history -> call the smallest useful tool -> reason with a focused agent -> return an answer or notification.
The dashboard remains useful without AI; AI enriches it with current, contextual, actionable guidance.

Demo Time





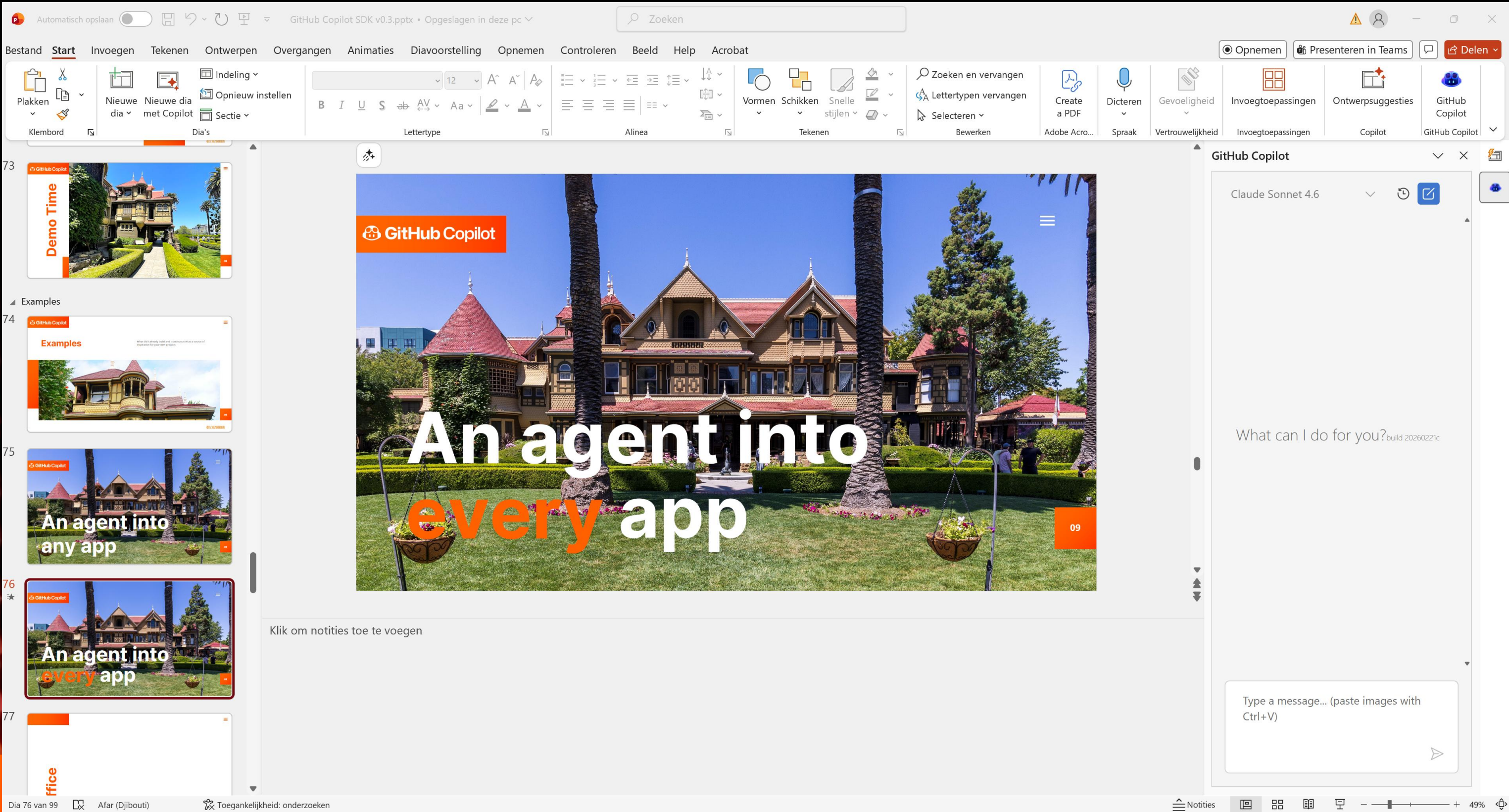
Examples

What did I already build as a source of inspiration for your own projects

An agent into any app

An agent into every app

Office Pilot



Getting started with the GitHub Copilot SDK in .NET

May 07, 2026

In the previous post talked about why the GitHub Copilot SDK matters: it gives you a production-grade agent harness out of the box, so you can skip building the infrastructure and focus on your actual product. Now let's make it concrete. This post walks through everything you need to get up and running with the SDK in .NET — from prerequisites to a working streaming agent with a custom tool.



What we will build

SHARE

Labels

.NET

.NET 10

AI

Github Copilot

GitHub Copilot Browser

Copilot Browser

Error

+ New

What can I help with?

Ask me to interact with the current page, fill forms, extract data, or navigate the web.

Ask Copilot...

Copilot can interact with the current page

Business
 Pilot


 Vlaanderen

Vlaamse Landmaatschappij

VLAAMSE LAND MAATSCHAPPIJ


 Vlaanderen
is open ruimte

StartStop beheer

StartStopLijsten

STARTSTOP BEHEER

Home

Beheer

Openstaande Startbrieven

Query's

Querybouwer

API referentie

OPGESLAGEN

+ Nieuw

Query's

- Toegang Mestbankloket OK = nee

x
- IdentOK = nee per PA

x
- Opvolgen niet leeg per PA

x
- Opvolgen niet leeg - alle PAs

x

AI-ASSISTENT

Schema opnieuw laden

Query genereren

Beschrijf welke data je wilt ophalen, bijv. Geef mij de eerste 10 landbouwers met status Stop...

Genereer

EDITOR

Opslaan

Uitvoeren

Naam van de query...


```

1 {
2   startStopRecords {
3     nodes {
4       id
5       landbouwersnummer
6       exploitantnaam
7     }
8   }
9 }
      
```

DEVPILOT



Code Reviewer



THE ORACLE



Conclusion

What should I take away from this session?

You **don't**
need to build
your own
harness



An agent into every app

Thank You

